

CAPSTONE PROJECT · 2026

Corporate Network **Infrastructure** Proposal

Shay Crane · Networking & Cybersecurity AAS
Independent Capstone Project





CLIENT

Medium-Sized Law Firm

Requires a full enterprise network infrastructure to support professional legal operations across all staff and departments.



DIGITAL TRANSITION

Beginning of Transition to Paperless Office Initiative

The firm is discontinuing all paper-based client files in the future and will be moving to a completely digital office — requiring robust, reliable infrastructure to support the transition once the corporate domain network proves itself valuable.



GOAL

Design & Deploy Enterprise-Grade Corporate Network

Build a scalable, secure, enterprise-class network to fully support the firm's digital operations — from domain infrastructure to remote access and cybersecurity.





01

50-User Domain

Corporate Windows Server domain for all staff — Active Directory, DNS, DHCP, and RRAS provisioned for the full firm.



02

Hyper-V Environment

Nested virtualization hosting all server roles — domain controller, file, print, and SQL servers — on Cisco UCS hardware.



03

Network Shares

Shared file server, print server, and SQL Server deployment — centralized storage and services accessible to all domain users.



04

Subnet Plan

Full IP addressing and subnet documentation for the 192.168.10.0/24 LAN — static reservations, DHCP scope, and VPN pool defined.



05

Security & Backup

Incident response plan, EDR, MFA, RBAC, firewall policy, and daily/weekly/monthly backup and recovery procedures deployed.



06

Budget & Timeline

Documented hardware and licensing cost estimates with a structured 6-week delivery schedule from planning through final presentation.



Segment	Network	Gateway	Mask
WAN	10.0.0.0/24	10.0.0.1	255.255.255.0
LAN (Corp)	192.168.10.0/24	192.168.10.1	255.255.255.0

Parameter	Value
Scope Name	CorpNet - LAN
Network	192.168.10.0/24
Lease Range	192.168.10.100 - .200
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
DNS Server	192.168.10.10
DHCP Server	192.168.10.10 (SRV-DC01)

- Router — WAN: 10.0.0.1
- Firewall — External: 10.0.0.2
- CorpNet Switch — LAN: 192.168.10.1
- SRV-DC01 — Int: 192.168.10.10 · Ext: 10.0.0.1
 - Roles: AD DS · DNS · DHCP · RRAS
- File Server — \\SRV-DC01\CorpData · logical



•
192.168.10.0/24



LAN subnet for all domain clients

•
.50 – .200



151 dynamic addresses for workstations & clients

•
.1 – .49



Reserved for static assignments (servers, infrastructure)

•
PC-01 – PC-50



Plus other dynamic LAN clients



→ SEQUENTIAL PROCESS — 4 STEPS

1

VPN TECHNOLOGY

RRAS PPTP VPN

Routing and Remote Access Service (RRAS) configured with PPTP VPN for authorized remote access to domain servers. Allows off-site users to securely reach internal resources.



2

IP ASSIGNMENT

Dedicated RRAS Pool

VPN users receive IP addresses from the dedicated RRAS static pool — entirely separate from the DHCP scope (.50–.200). This keeps VPN and LAN address spaces isolated and avoids conflicts.



3

FIREWALL POLICY

Controlled Traffic Rules

Firepower rules allow PPTP inbound for VPN only. All direct external access to network devices is blocked. Outbound web traffic is filtered through dedicated outbound firewall rules.



4

ACCESS CONTROL

RBAC — Remote Access Users

Only members of the Active Directory RBAC group "**Remote Access Users**" may connect via VPN. Valid domain credentials are required to authenticate — no guest or unauthenticated sessions permitted.





SECURITY PLAN



PERIMETER DEFENSE

Perimeter Firewall

Cisco Firepower 1010 controls all inbound and outbound network traffic at the edge.



ACCESS CONTROL

RBAC

Role-Based Access Control implemented via Active Directory; access scoped by job role.



ENDPOINT DETECTION

EDR

Cisco Secure Endpoint: lightweight agent for malware prevention, threat detection, and event logging.



MULTI-FACTOR AUTH

MFA

Cisco Duo MFA for 100 users; adds a second verification layer to every domain login.



THREAT PREVENTION

IPS & URL Filtering

Cisco Firepower subscription for intrusion prevention and outbound URL filtering across all users.



BACKUP & RECOVERY



PROTECTED SERVICES

All critical domain services must be recoverable.

AD DS

DNS

DHCP

RRAS

File Server

Print Server

SQL Server



SERVER RECOVERY

Restore System ›

Reapply Service Configs ›

Recover Shared Roles ›

Confirm Domain Functional



DATA RECOVERY

Restore File Shares & SQL DB/Logs ›

Validate Access Controls ›

Confirm Data Integrity



BACKUP FREQUENCY

Daily — Incremental

Weekly — Full Backup

Monthly — Archive



CLIENT RECOVERY

Re-flash Workstations ›

Rejoin Domain ›

Restore User Profile & Data

PROJECT BUDGET

Scale: \$0 – \$64,100 (Year 1 High)

Hardware

Cisco Firepower 1010 · Catalyst 9300 · Catalyst 9200 · 150AX AP · UCS C220 Server

LOW

\$16,700

HIGH

\$37,300

Licensing

Cisco Duo MFA · Cisco Secure Endpoint · Firepower IPS / URL Filtering / AMP Subscription

LOW

\$10,100

HIGH

\$26,800

\$ Grand Total — Year 1

Combined hardware + licensing investment for full enterprise deployment

LOW

\$26,800

HIGH

\$64,100

YEAR 1 INVESTMENT RANGE

\$26,800

\$64,100_{Low–High Estimate}

Includes all hardware & software licensing. Final cost depends on vendor pricing & configuration scope.

HARDWARE ITEMS

Cisco Firepower 1010	Included
Cisco Catalyst 9300	Included
Cisco Catalyst 9200	Included
Cisco 150AX Wi-Fi 6 AP	Included
Cisco UCS C220 Server	Included

LICENSING ITEMS

Cisco Duo MFA	Included
Cisco Secure Endpoint (EDR)	Included
Firepower IPS / URL / AMP	Included



IMPLEMENTATION TIMELINE

🔍 Planning

Requirements gathering
Network design
IP addressing plan
Budget & device selection

wk 1

wk 2

wk 3

wk 4

wk 5

wk 6

🔌 Infrastructure Setup

Firewall config
Router & switch setup
Wireless AP integration
Subnet validation

💻 Client Deployment & Testing

PC-01 to PC-50 domain join
GPO validation
Connectivity testing

💻 Virtualization & Server Deploy

Hyper-V setup
SRV-DC01 deployment
AD DS / DNS / DHCP / RRAS

🛡️ Security Configuration

EDR deployment
Firewall policy tuning
RBAC & MFA integration
Incident response docs

📄 Final Documentation

Network diagram
Security & budget docs
Backup / recovery plan
Final presentation

6-WEEK DELIVERY

● Milestones ○ Completion

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Thank You

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